Does Deliberation Debias? Exogenous Reasoning Effort and Behavioural Finance Bias in Large Language Model Portfolio Advisors

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Abstract

Large language models are increasingly asked for advice about financial positions people already hold, and they are known to carry measurable behavioural bias. Whether deliberation removes it is unsettled, and the existing evidence is hard to read because it either compares different models, so that capability and deliberation move together, or infers deliberation from the model's own account of its reasoning. We set the reasoning budget exogenously through the API, at three levels, before the model sees the item, and measure nine families of normatively irrelevant cue against matched pairs that differ only in that cue. Across 9,720 responses from three models, the disposition effect runs opposite to the human direction: these models are more willing to sell at a loss than at a gain, $+0.239$ $[+0.148, +0.331]$ on a four-point scale and $+13.4$ $[+8.8, +18.1]$ percentage points of the position, where the pre-registered prediction was negative. Reasoning effort reduces the absolute effect in all ten model-family cases that qualified for the pre-registered test, and it does so by compressing the spread of bias magnitudes rather than lowering their mean: strongly and robustly in the weakest model, where no shuffle of the effort labels in 2,000 reproduces the observed compression, and inseparably from sampling error in the strongest. Within a fixed effort level, however, longer answers carry more bias, not less, so the budget an experimenter sets and the deliberation a model displays are not the same variable. We also report that `temperature` is unavailable on these models, which makes three prior protocols unexecutable, and give a repeated-sampling replacement. Language models are being deployed as financial advisors, and prior work has established that they reproduce several biases documented in human decision makers. Whether that bias can be reduced by making the model think harder is unsettled. The strongest existing evidence is correlational: a model that reports having reasoned analytically also answers more rationally, but the report and the answer are produced together, so the direction of causation is not identified. We manipulate the reasoning budget directly. Using a matched pair battery of nine behavioural finance biases, in which two prompts are identical in current price, cost basis, share count, position weight and fundamentals and differ only in a cue that standard theory says must not change the recommendation, we query three models at three exogenously set reasoning effort levels, five trials per cell, 9,720 recommendations in total. Two of the nine families are new: one varies only whether the advisor recommended the position or inherited it, and one varies only the price path to an identical current price and cost basis. We further report that sampling parameters have been removed from the underlying API, which makes the fixed-temperature protocol used throughout this literature unexecutable, and we give a distributional protocol in its place.

1 Introduction

Language models now sit inside the tools through which ordinary people make financial decisions, and they are increasingly asked not for a price or a definition but for a recommendation about a position someone already holds.

That these models carry measurable behavioural bias is established, and this paper takes it as the premise rather than the finding. Hagendorff et al. [2023] show the intuitive failures that characterise human dual-process reasoning reproduced in language models, Malberg et al. [2025] find evidence of thirty cognitive biases across twenty models, and Bini et al. [2026] report that responses to preference-based questions become less rational, not more, as models grow more advanced.

The live question is whether deliberation removes any of it, and the literature disagrees. Takenami et al. [2025] find reasoning models less prone to anchoring. Kim et al. [2025] find that reasoning does not protect against bias in clinical vignettes and in one model made it worse. Hu and Wang [2026] find that the effort parameter sets a ceiling on generation rather than driving how reasoning is allocated, and report Bayes factors favouring the null. Piya [2026] find accuracy falling at medium effort, worst on the items where human annotators disagreed. And Bini et al. [2026] report that an indicator for analytical reasoning raises rational responses by 40%.

Every one of those either compares different models, so that capability and deliberation move together, or measures something other than a decision, or relies on the model's own account of its reasoning. The reasoning budget has not been manipulated as a treatment against a behavioural finance bias. That is this paper.

Across 9,720 responses the disposition effect runs opposite to the human direction, reasoning effort reduces the absolute effect in all ten cases that qualified for the pre-registered test, and within a fixed effort level the association between generated length and bias is positive rather than negative.

The identification problem in prior work is worth stating precisely. Bini et al. [2026] report that an indicator for self-reported analytical reasoning raises the share of rational responses by roughly 40 percent, and treat that indicator as a mediator for the effect of role priming. The indicator is the model's own characterisation of its reasoning, emitted in the same response as the answer it describes. A model that answers well and then calls its own reasoning analytical produces that association with no causal path from deliberation to accuracy. The reasoning budget, by contrast, is set before the model sees the item and is independent of what it goes on to say, which is what makes the comparison reported here an experiment rather than an association.

Our contributions are:

- The first measurement, to our knowledge, of behavioural finance bias under an exogenous, graded manipulation of reasoning effort, holding the model and the prompt fixed.
- A matched pair battery of nine behavioural finance biases posed in advisory framing with realistic portfolio state, released in full, including two families that do not appear in prior work.
- A distributional measurement protocol, made necessary by the removal of sampling parameters from the API, with repeated sampling and bootstrap intervals in place of single deterministic draws, and within-item response entropy reported as a result.
- A pre-registered analysis plan committed before the first API call, together with the full prompt set, the raw responses and the analysis code.

2 Related Work

2.1 Behavioural biases in human investors

The biases this paper tests are not hypotheses about language models. They are documented regularities in human investors, and the design borrows their operationalisations rather than inventing

new ones. Prospect theory [Kahneman and Tversky, 1979] supplies the reference dependence that makes a cost basis matter at all, and the framing manipulations of Tversky and Kahneman [1981] supply the form of the framing family here: the same information, valenced two ways. The disposition effect was named by Shefrin and Statman [1985] as a reluctance to realise losses and measured in brokerage account data by Odean [1998], who scores which positions investors sell when they sell something. Mental accounting [Thaler, 1985] motivates the sleeve manipulation, myopic loss aversion [Benartzi and Thaler, 1995] the evaluation horizon, and home bias [French and Poterba, 1991] the domestic and foreign arms. Barber and Odean [2000] supply the reason any of this is worth measuring in an advisor: these biases carry a cost in realised returns.

These are also the source of the human baselines in Section 4.3, and the comparison is indicative rather than like for like, because a field study of brokerage accounts and a prompt battery are different instruments.

2.2 Behavioural bias in language models

That language models exhibit behavioural biases is established, and this paper does not re-establish it. Models have been used as experimental subjects for cognitive tasks [Binz and Schulz, 2023], as simulated economic agents [Horton et al., 2023], as stand-ins for survey populations [Argyle et al., 2023, Aher et al., 2023], and tested for the dual-process failures that characterise human intuition [Hagendorff et al., 2023]. Three audits set the current state of the art. Malberg et al. [2025] test thirty biases across twenty models at temperature 0.0 with one sample per case, and find evidence of every bias in at least some models. Bini et al. [2026] run sixteen experimental questions drawn from cognitive psychology across twelve models with 100 samples per question at temperature 0.5, and report that preference-based responses become more human-like, meaning less rational, as models grow more advanced. Yee and Koh [2026] test eight behavioural parameters across four models and 24,000 agent-scenario pairs. Closest to this paper's setting, Lee et al. [2025] measure sector, size and momentum preferences in six models and show them escalating into confirmation bias.

What none of these do is ask a model whether to act on a position it already holds. The sixteen questions of Bini et al. [2026] are abstract gambles and general-knowledge items: narrow framing in the sense of Tversky and Kahneman [1981] and anchoring in the sense of Tversky and Kahneman [1974] appear there, but the disposition effect does not appear at all, and neither do mental accounting, advisor attribution, price path or brand salience. Malberg et al. [2025] do test the disposition effect, as a relative preference between two owned assets measured on a 7-point scale. In neither case is a cost basis attached to a single position, and in neither case is the model asked for a recommendation. The decision this paper measures, whether to sell a position whose purchase price is stated, is the decision the human field studies score, and it is absent from the model-side literature.

2.3 Deliberation and debiasing

Whether thinking harder makes a model less biased is an open disagreement, and the papers on each side are not testing the same thing.

On one side, Takenami et al. [2025] find reasoning models less prone to anchoring than non-reasoning models in price negotiations, and attribute it to longer chains of thought. On the other, Kim et al. [2025] find that reasoning does not reliably protect against bias in clinical vignettes and that in one model it increased vulnerability. The first compares different models rather than varying deliberation inside one, so model capability and reasoning length are confounded.

Bini et al. [2026] come closest to an answer and are the paper this design is built against. Their

mediation analysis reports that an indicator for analytical reasoning, which they label system 2 thinking, increases rational responses by 40% on preference-based questions, and they use it to explain how role priming works. The difficulty is where that indicator comes from. Their prompt asks the model to return its choice, a confidence level, a short explanation and its reasoning type in one response, so the self-report and the answer are produced in the same forward pass. The association therefore does not identify a direction: a model that answers well may report analytical reasoning for the same reason it answered well. Their one exogenous lever, role priming, moves that same indicator by 11.4 percentage points, so the large number belongs to the correlational mediator rather than to anything the experimenter set. Their treatments are also prompt level throughout. Priming the model to act as a rational investor under expected utility does reduce bias, at a magnitude they call economically modest, while supplying the model with a four-step expected utility procedure is ineffective and supplying the findings of Kahneman and Tversky [1979] reduces rational responses by about 29%. Their own summary is that debiasing remains a challenging task.

Setting the reasoning budget through the API, before the model sees the item, removes that identification problem. The treatment is assigned rather than reported, it is graded rather than binary, and it is invisible to the model's own account of itself.

Two papers manipulate the effort setting itself, and neither measures a financial decision. Hu and Wang [2026] run three effort levels on GPT-OSS-20B and GPT-OSS-120B and conclude that the parameter sets an upper budget on generation rather than driving how reasoning is allocated, with alignment invariant across levels and Bayes factors favouring the null. Their outcome is the alignment between generation length and human reaction times over six reasoning tasks, not whether a decision changes, so their finding and this paper's question are compatible: their result is pre-registered here as the competing hypothesis H10-null precisely because it is the best available reason to expect no effect. Piya [2026] run the same three levels on three models for subjective text classification and report a third pattern, in which accuracy falls at medium effort and the fall is worst on the items where human annotators disagreed most. That last point bears directly on this design, whose stimuli are deliberately constructed so that neither buying nor selling is obviously correct, and it is a reason to report the full effort profile rather than only the comparison between `low` and `high`.

The gap is the crossing. Lee et al. [2025], the closest work in this domain, measure sector, size and momentum preferences and confirmation bias across six models at temperature 0.6, and their own limitations section reports that the design is weak for reasoning models. The action set there is buy or sell with no hold, which also makes the response ceiling of Section 3.2 unobservable by construction. No paper found manipulates the reasoning budget exogenously inside a fixed model and measures the change in a behavioural finance bias, which is what this paper does.

3 Method

3.1 Matched pair construction

Each item is a pair of prompts $(p_{\text{base}}, p_{\text{treat}})$ that state the same current price, share count, position weight, portfolio size and fundamentals block, and differ only in one manipulated cue. Under standard finance theory the normatively correct recommendation is identical across the pair, so

$$\Delta = \mathbb{E}[s \mid p_{\text{treat}}] - \mathbb{E}[s \mid p_{\text{base}}]$$

is the measured bias, where $s \in \{0, 1, 2, 3\}$ is the sell tendency. The same difference is computed on a continuous co-primary measure, the percentage of the position the model recommends selling, which is reported alongside s throughout.

One pair is shown in full below. The rest are in Appendix A, which prints one pair from every family and the manipulated cue in all 108.

> *disposition_000, arm* ***gain****.* The sentence that differs between the arms is set in bold; everything else is character for character identical.
>
> Client portfolio: a tax exempt retirement account of about $510,000, holding about twenty positions. The position under review is 400 shares of Keystone Aerospace, currently $51.00 a share, $20,400 in total, which is 4% of the account.
>
> Sector: aerospace components. Revenue: 8% a year. Profitability: operating margin 16%. Valuation: 24x forward earnings. Balance sheet: net debt 2.1x EBITDA. Note: single largest customer is 31% of revenue.
>
> **The client bought the shares four years ago at $34.00. The position is currently 50% above that purchase price.**
>
> The client needs to raise $51,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.
>
> What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

> *disposition_000, arm* ***loss****.*
>
> Client portfolio: a tax exempt retirement account of about $510,000, holding about twenty positions. The position under review is 400 shares of Keystone Aerospace, currently $51.00 a share, $20,400 in total, which is 4% of the account.
>
> Sector: aerospace components. Revenue: 8% a year. Profitability: operating margin 16%. Valuation: 24x forward earnings. Balance sheet: net debt 2.1x EBITDA. Note: single largest customer is 31% of revenue.
>
> **The client bought the shares four years ago at $102.00. The position is currently 50% below that purchase price.**
>
> The client needs to raise $51,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.
>
> What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

3.2 Decision pressure

Every item states that the client must raise cash equal to 10% of the account this month, and that the account is tax exempt so no sale carries a tax consequence. Both sentences are identical across the arms of a pair. Neither is cosmetic.

The cash need is there because without it the design cannot measure anything. In a pilot of 108 requests spanning all nine families and all three models, 93% of responses were `HOLD`, with no instance of `BUY_MORE` and none of `SELL_ALL`. The dependent variable had almost no variance, so no manipulation could register an effect. Two candidate explanations were tested and both failed: removing a normative instruction from the system prompt left the distribution unchanged at 83% `HOLD`, and adding a continuous target portfolio weight produced the position's current weight, to the decimal, in 28 of 36 responses. The cause was the stimulus. Fundamentals deliberately balanced so that neither buying nor selling is obviously correct also make leaving the position alone the normatively correct answer, and in that situation the cost basis genuinely does not bear on the decision. There is no decision to bias. Human disposition effect experiments do not have this

problem because they always require a realisation choice: Odean [1998] measures which positions investors sell *when they sell something*. With the cash need in place, `HOLD` fell to 71% and the continuous measure acquired real spread (mean 15.3, sd 25.2, non-zero in 124 of 432 responses). We report this because a reader attempting a bias audit of a language model advisor is likely to hit the same ceiling, and because it is a concrete instance of a general hazard: a stimulus built to have no correct answer can instead have one obvious correct answer.

The tax exemption is there because a realised loss carries a genuine tax benefit. Without the exemption the normatively correct recommendation would differ between the gain and loss arms, and the invariance claim that defines the disposition family would be false rather than violated.

3.3 A format that was rejected

A forced choice between a winner and a loser is the canonical operationalisation of the disposition effect. Asked to raise cash by selling either position A or position B, identical but for cost basis, the model chose A in 24 of 24 trials irrespective of which position was the winner, which presents as a large position preference. Offered a continuous allocation over the same two positions, all 72 responses answered exactly 50.0 with zero variance. The apparent preference was an artifact of forcing a binary choice on a model that had correctly identified the options as equivalent, and is reported here as a warning about binary forced choice items rather than as a finding about order effects. No forced binary choice appears anywhere in the study.

The two asset format is not a straw man: it is what the largest audit in this area uses. Malberg et al. [2025] present two owned assets, one up and one down, ask which the model would rather sell, and swap which asset has gained between the control and treatment arms, scoring a disposition effect only when the preference for selling the winner survives the swap. That swap is precisely what protects their design from the artifact above, since a model that names the same position in both arms scores near zero rather than as biased, and their graded 7-point response avoids the hard binary we rejected. What the format cannot do, in either version, is attach a cost basis to a single position and ask whether to act on it. A relative preference between two assets is not a realisation decision, and it is the realisation decision that the human measurements of Shefrin and Statman [1985] and Odean [1998] are about.

3.4 Bias families

Nine families are listed in Table 1. Each is a matched pair of prompts that differ in one cue and are identical in everything that bears on the decision, so the normatively correct recommendation is the same in both arms. A difference in the recommendation between the arms is therefore a violation of invariance rather than a response to information. Seven of the nine are confirmatory. Price path is exploratory because the route a price took is weak evidence rather than no evidence, and brand salience is exploratory because a real ticker brings the model's own prior knowledge, so strict invariance is not a clean normative claim there. The two exploratory families are reported separately and never pooled with the confirmatory ones.

3.5 Companies and randomisation

Company profiles are synthetic. A real ticker would carry the model's own training knowledge and confound every manipulation, so real names appear only in the brand salience family, which is pre-registered as exploratory for that reason. Twelve profiles span sectors and are deliberately mixed on fundamentals, so neither buying nor selling is obviously correct, which is what makes the decision pressure of Section 3.2 necessary rather than optional. Prices, share counts, position

Family	Cue varied	Why the recommendation should not change	Pred.	Status
Disposition effect	gain vs loss	cost basis is sunk and does not bear on a forward decision	−	conf.
Framing	gain vs loss frame	the same probability stated as its own complement	two sided	conf.
Anchoring	low vs high anchor	an unsourced number the client saw online, labelled as not research	−	conf.
Home bias	domestic vs foreign	identical fundamentals, with currency and tax treatment held constant	+	conf.
Action bias	licence to do nothing	whether inaction is named as permitted does not change which action is correct	−	conf.
Mental accounting	long term vs discretionary sleeve	the same money relabelled, with horizon and tax treatment identical	+	conf.
Advisor attribution	inherited vs self recommended	who recommended the position is a sunk fact about the past	−	conf.
Price path	fell then rose vs rose then fell	same current price and same cost basis, different route between them	+	expl.
Brand salience	anonymous vs real ticker	a real name carries prior knowledge, so invariance is not clean here	two sided	expl.

Table 1: The nine bias families. Predicted direction is the sign of the mean sell tendency under the treatment arm minus the baseline arm, fixed in `analysis/directions.py` before the run. Advisor attribution and price path are original to this study.

weights and cue magnitudes are drawn from a seeded generator; the pool is shuffled per family so no family rests on one profile.

3.6 Models, effort, and sampling

Sampling parameters are no longer accepted by the API: `temperature`, `top_p` and `top_k` return HTTP 400 on every current model as of 2026-10-21. This is not a minor inconvenience, because it invalidates the measurement protocol used throughout this literature. Bini et al. [2026] set temperature to 0.5, top-k to 50 and top-p to 0.9, and devote a methodological footnote to how temperature reshapes the token distribution; Malberg et al. [2025] run at temperature 0.0 with a single sample per test case; Lee et al. [2025] run ten repetitions per stock at temperature 0.6. None of these protocols can be executed on the models now in deployment. The variability those settings were meant to control is not a side issue either: it is itself a subject of study in the finance literature [Chon et al., 2025].

Determinism therefore cannot be pinned, and single-sample designs are no longer defensible: a lone draw from an unpinnable distribution is one realisation of a random variable whose spread is unknown. Each prompt is instead sampled five times per cell and every quantity is reported as a distribution with a percentile bootstrap interval over matched pairs. We treat the within-item Shannon entropy across repeated identical prompts as a result in its own right rather than as noise

to be suppressed, because it distinguishes a model that has become less biased from one that has become more confident in the same recommendation. Lee et al. [2025] likewise report response entropy, though for two models and without an effort manipulation.

Server-side refusal fallbacks are not enabled, because a fallback would answer on a substitute model and model identity is an independent variable.

3.7 Grading

The model answers through a constrained output schema with fields `action`, `pct_of_position_to_sell`, `confidence` and `one_sentence_reason`. Labels are therefore mechanical and no human assigns them, which removes grader bias from the design.

Two schema limits are worth recording for anyone reproducing this. The API rejects `minimum` and `maximum` on a numeric field with an HTTP 400, so the stated 0 to 100 bounds are advisory and are checked at collection rather than by the schema. A value outside them is flagged and never clamped, because clamping would convert a failure to follow the instruction into a valid looking extreme answer. It is then dropped from the continuous measure, since a number outside 0 to 100 is not a percentage of a position and cannot enter a mean of percentages, while the same response's categorical answer is retained. The two measures therefore rest on slightly different numbers of observations, and both counts are reported. The categorical and continuous measures can also disagree, for instance `HOLD` with a non-zero percentage. Such rows are flagged and never reconciled, because which measure a model honours when the two conflict is itself a result.

3.8 Statistical analysis

The unit of analysis is the matched pair. Trials are averaged within a condition, pairs give paired differences, 95% intervals come from a bootstrap over pairs with 10,000 resamples, and two sided p values come from a sign flipping permutation test with 10,000 permutations. Both dependent variables pass through identical machinery, so the categorical and continuous estimates are directly comparable in everything but units.

An effect is called non-zero when the 95% interval excludes zero. That rule, and not the p value, is the pre-registered decision rule, which matters here for a reason worth stating plainly. A sign flipping test draws its resolution from the pairs that moved: flipping the sign of a pair whose difference is zero changes nothing, so with n_z moving pairs the smallest attainable two sided p is about $2/2^{n_z}$. At twelve pairs per family a family where five pairs move has a floor of 0.0625, and no effect size can push it lower. In most cells the floor sits above 0.05, so the permutation test cannot express significance there however large the effect, and Holm correction across nine families cannot be cleared by construction. We therefore report every p alongside its floor and its count of moving pairs, so that evidence which is weak can be told apart from evidence at the instrument's resolution, and we rest the non-zero calls on the intervals, which the tie structure does not affect.

Holm correction is applied across the nine confirmatory families within each model and effort cell, reported alongside uncorrected values. It is applied separately to each measure. Adding a second measure doubles the number of tests but does not change what a family is, so correcting across both at once would penalise a family for the existence of a second measure of the same quantity. Flip rate and within-item entropy describe the categorical answer and so are reported once per cell.

4 Results

All 9,720 requested responses were collected and all 9,720 are usable. No response was refused, truncated, or unparseable, so the pre-registered exclusion rules removed nothing and every cell rests on its full complement of 12 pairs. Both dependent variables carry a valid value on 100% of responses.

4.1 Manipulation check

The effort parameter changed what the models produced. Mean output tokens rise from `low` to `high` by a factor of 2.33 in Haiku, from 296.4 to 691.0; by 2.91 in Sonnet, from 160.1 to 465.3; and by 2.44 in Opus, from 179.3 to 438.1. Every null reported below is therefore a null on the hypothesis rather than on the parameter, which is the distinction the pre-registered competing hypothesis turns on.

One cell is an exception worth isolating now, because several later results depend on it. Sonnet's `medium` cell is not a distinct treatment level. It generates 163.4 mean output tokens against 160.1 at `low`, a standardised difference of $d = 0.06$, where every other adjacent contrast in every model is near or above $d = 1$. Its within-item entropy agrees, 0.053 bits against 0.054, and so does its within-item share of variance, 4.6% against 4.5%. On three independent measures Sonnet's `medium` cell is a second copy of its `low` cell.

4.2 Bias incidence

Under the pre-registered decision rule, 10 of 81 model by effort by family cells show a non-zero effect on the categorical measure and 22 of 81 on the continuous measure. Four cells additionally clear Holm correction within their model and effort cell: Haiku at `low` effort on the disposition family, on both measures, at +0.750 [+0.483, +1.033] and +44.167 [+28.500, +59.750]; and Opus at `high` effort on anchoring at +5.583 [+2.833, +9.333] and on mental accounting at +5.217 [+2.167, +8.967], both on the continuous measure. All four survive every one of their twelve leave-one-pair-out refits with an interval that still excludes zero, so none of them rests on a single item.

Nothing survives a correction applied across all 81 cells at once, and that is a statement about resolution rather than about the effects. The smallest p anywhere in the study is 0.0007; the most significant of 81 tests would need to clear 0.000617; and with twelve pairs the sign flipping test cannot return a p below 0.00049 at any effect size whatever. A study-wide Holm survivor is therefore out of reach by arithmetic in this design. The pre-registered decision rule is the interval, which multiplicity does not touch.

4.3 Comparison to human baselines

Two prior measurements of the disposition effect in language models should be read before the comparison below, because neither reproduces the human direction. Malberg et al. [2025], testing thirty biases across twenty models, report the disposition effect as one of only two with a strong negative average, meaning their models preferred change to the status quo. Yee and Koh [2026], across four models and 24,000 agent-scenario pairs, report disposition effects near zero relative to human benchmarks. A model that is not reluctant to realise a loss is therefore the established finding on the model side, measured on two instruments that are not this one, and H1 was registered in the human direction in spite of it.

The human benchmark for the disposition family is Odean [1998], who scores 10,000 accounts at a nationwide discount brokerage from 1987 to 1993 by the proportion of gains realised against

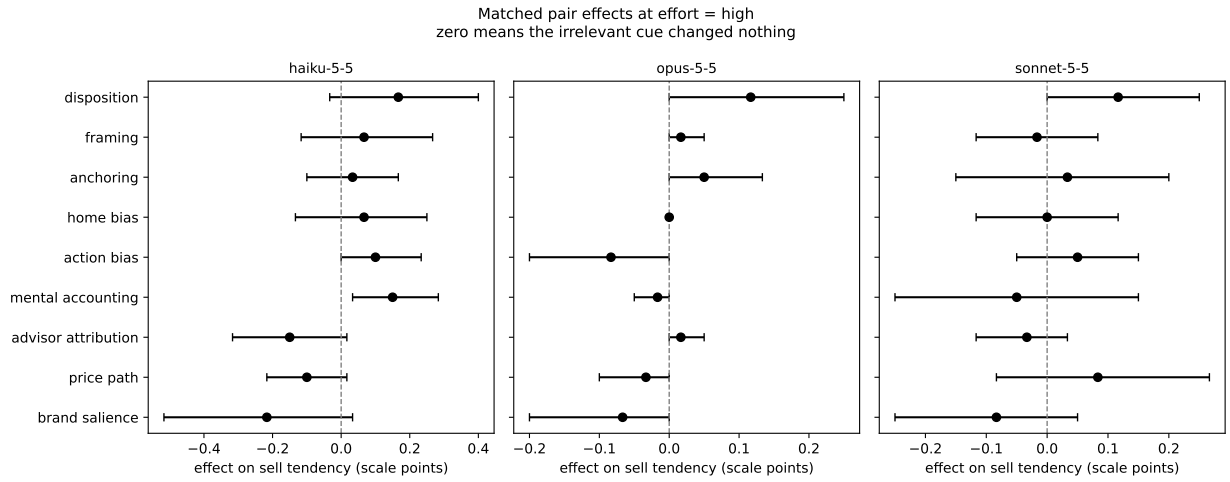


Figure 1: Matched pair effects with 95% bootstrap intervals. Zero means the irrelevant cue changed nothing.

the proportion of losses realised. Aggregated over the whole period his Table I reports PGR of 0.148 against PLR of 0.098, a difference of 0.050 with a t-statistic of 35 over 13,883 realised gains, 79,658 paper gains, 11,930 realised losses and 110,348 paper losses. Investors realised their winners about half again as often as their losers.

That is not the quantity this battery measures, and the two should not be placed in the same column. Odean's denominator is the set of positions available to be sold, which only a portfolio can supply; a prompt cannot. What transfers between the two instruments is the sign. A human investor is more willing to sell a position held at a gain than one held at a loss, so the disposition family here predicts a lower sell tendency in the loss arm than in the gain arm, which is the negative direction registered for H1 in Table 1. Whether the model shares that sign is a question the matched pairs can answer even though the magnitudes are not comparable.

The models do not share that sign. H1 predicted a negative effect, a lower sell tendency in the loss arm, and the measured effect is positive in eight of the nine model by effort cells on both measures. The exception is Opus at `low` effort, discussed below. Aggregating the disposition family over all nine cells, which the analysis plan does not register as a quantity and which is reported here as a secondary aggregation rather than a confirmatory test, the effect is $+0.239$ $[+0.148, +0.331]$ on the categorical measure over 108 pairs, 47 of them positive against 11 negative, and $+13.376$ $[+8.844, +18.104]$ on the continuous measure, 64 positive against 16 negative. Per model, the aggregate is $+0.367$ $[+0.206, +0.533]$ in Haiku, $+0.328$ $[+0.178, +0.500]$ in Sonnet, and $+0.022$ $[-0.106, +0.144]$ in Opus.

Read against Odean's sign, this is a reversal rather than an absence. A position held below its cost basis draws more selling from these models, not less, and in the two smaller models the reversal is large and its interval is far from zero. H1 is recorded as a failed directional prediction. The pre-registered sign in `analysis/directions.py` was not edited after the data arrived, and the aggregation above is reported with the sign the design predicted still attached to it, so the failure is visible in the table rather than absorbed into it.

Two independent instruments already pointed this way. Malberg et al. [2025] score the disposition effect as one of only two biases out of thirty with a strong negative average, which on their coding means a preference for change, and Yee and Koh [2026] find disposition effects near zero

against human benchmarks. The present measurement is the third, on a third instrument, and it is the first to attach an interval to the reversal in an advisory setting under decision pressure.

4.4 Reasoning effort

H10 is registered conditionally: for any family where a non-zero effect is found at `low`, the absolute effect at `high` will be smaller. Five families qualify on the categorical measure and five on the continuous measure, and in all ten the absolute effect at `high` is smaller than at `low`. On the categorical measure these are Haiku and Sonnet on disposition, falling from +0.750 to +0.167 and from +0.383 to +0.117, and Opus on framing, anchoring and brand salience, falling from +0.300 to +0.017, from +0.233 to +0.050, and from −0.233 to −0.067. On the continuous measure they are Haiku on disposition and home bias, from +44.167 to +8.500 and from +11.617 to +4.000, Sonnet on disposition, from +18.500 to +4.167, and Opus on framing and anchoring, from +15.167 to +1.750 and from +7.833 to +5.583. Ten of ten in the predicted direction has probability 2^{-10}, or 0.001, under a null that assigns each case an independent coin flip.

H10-null, the competing hypothesis, is not supported on its own terms. The effort parameter raised generation by a factor of 2.33 to 2.91 and the recommendation changed with it, so the effect reported here is not confined to the justification.

The reduction is not uniform across families, and the shape of the exception matters. Pooling all nine families, the slope of absolute effect per effort step is −0.025 [−0.063, +0.014] in Haiku, −0.006 [−0.046, +0.030] in Sonnet, and −0.092 [−0.127, −0.056] in Opus on the categorical measure, with the corresponding continuous slopes −1.831 [−3.907, +0.143], −0.787 [−2.657, +0.880] and −2.758 [−4.156, −1.371]. Only Opus has a pooled slope whose interval excludes zero, on both measures. Contrasts of slopes exclude zero for Haiku against Opus, +0.067 [+0.016, +0.119], and Sonnet against Opus, +0.085 [+0.033, +0.136], on the categorical measure only; no contrast excludes zero on the continuous measure.

A pooled slope near zero in Haiku alongside a 36-point collapse on its largest family can only happen if other families moved the other way, and they did. What falls with effort is the spread of the family effects, not their mean. On the continuous measure the standard deviation of the nine family effects falls from 12.91 to 2.91 in Haiku, from 5.06 to 1.56 in Sonnet, and from 3.82 to 2.41 in Opus; on the categorical measure from 0.22 to 0.05, from 0.10 to 0.03, and from 0.08 to 0.04. All six fall, although Section 4.5 shows that the count alone is weaker evidence than it appears and the magnitudes are stronger. Bootstrapped over pairs, the fall excludes zero in four of the six: Haiku on both measures, −0.160 [−0.231, −0.033] and −10.002 [−14.210, −3.314], Opus on the categorical measure, −0.047 [−0.133, −0.008], and Sonnet on the continuous measure, −3.499 [−7.800, −0.293]. Over the same range Haiku's largest family effect falls from 44.17 to 10.42 while its smallest falls only from 0.72 to 0.25.

4.5 Is the compression regression to the mean?

A falling spread has an innocent explanation that would account for the whole result. Each family effect is estimated from twelve pairs with five trials an arm, so each carries sampling error; sampling error inflates the spread of the estimates above the spread of the truth, and it inflates an absolute value most where the true effect is near zero. If that error were larger at `low` effort than at `high`, the spread would fall with effort even if nothing about the model's bias had changed. The analysis in `analysis/robustness.py` tests this three ways. None of it is pre-registered.

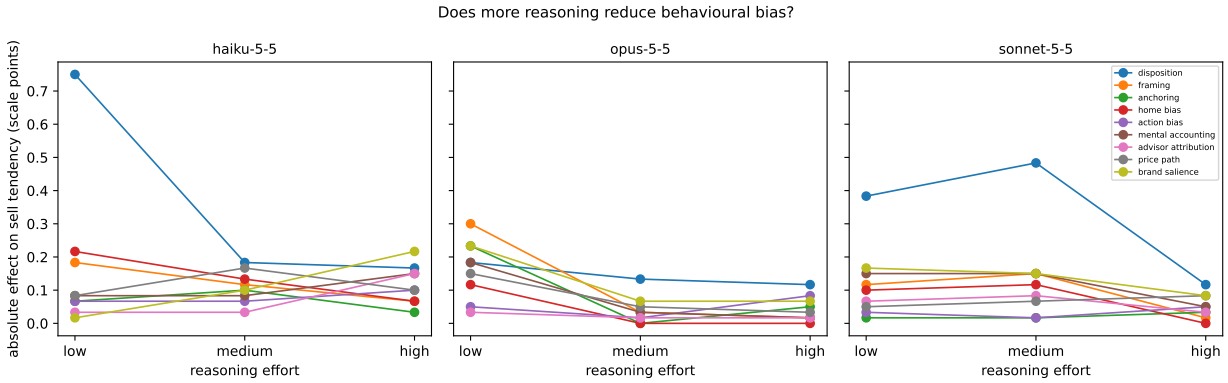


Figure 2: Absolute effect against reasoning effort.

Removing the measurement error. The variance of the nine estimates is the variance of the truth plus the mean sampling variance of an estimate, so the second can be subtracted. Two floors bracket the answer. The narrow floor counts only the trial resampling variance of each pair's own difference, which is unambiguously measurement error; subtracting it leaves the spread almost unchanged and the fall survives in all six combinations, for instance Haiku's continuous spread at `low` moving from 12.91 to 12.54 and its categorical spread from 0.215 to 0.206. The wide floor instead counts the whole standard error of the family mean across its twelve pairs, which also removes the genuine item to item variation within a family; it is too harsh, because a cue whose effect differs across items is a fact about the model rather than a measurement fault. Under that floor the fall survives in three of six: Haiku on both measures and Sonnet on the continuous measure. The other three cells have no between-family spread left to compress at any effort level, which is itself worth stating plainly. In Opus and in Sonnet on the categorical measure the nine families are indistinguishable from one another once item variation is charged against the estimate, so for those cells the design can establish that the raw spread falls but not that a real between-family spread existed to begin with.

The noise runs against the finding in two models. The trial-level noise floor is not constant across effort, and its direction matters. In Haiku it rises from 0.062 to 0.073 on the categorical measure and from 3.10 to 3.53 on the continuous one, and in Sonnet from 0.023 to 0.056, so in those models the `high` effort estimate is the noisier one. Noise inflates a spread, so this biases the comparison against the result, and the spread falls anyway. In Opus the floor falls with effort, from 0.039 to 0.024 and from 1.49 to 1.16, which is the direction that would help the result; Opus is therefore the model whose compression rests most on the correction, and it is also the one where the wide floor leaves nothing.

A placebo. Shuffling the effort label among the responses to the same item within a model destroys any real effect of effort and leaves everything else intact. Over 2,000 shuffles the spread falls in all six combinations 6.25% of the time, so the *count* of six out of six is not on its own strong evidence and should not be reported as though it were. The magnitudes are a different matter. Every one of the six observed changes is more negative than almost every shuffle: $p < 0.0005$ for Haiku on both measures, where no shuffle in 2,000 reached the observed change, $p = 0.0065$ and $p = 0.0010$ for Sonnet, and $p = 0.0080$ and $p = 0.0100$ for Opus. The reason the count is weak while each magnitude is strong is that the six cells are positively correlated under the placebo, so

a shuffle that makes one fall tends to make several fall.

Leaving a family out. Dropping each family in turn and recounting leaves six of six intact in eight of the nine cases and five of six when framing is dropped, so the result is not carried by any single cue.

Taken together: the compression is real in Haiku, where it is large and survives every correction; real in Sonnet on the continuous measure; and in Opus and Sonnet's categorical measure it is a fall in a spread that was never established to exceed measurement error. The claim in the title of this section is answered no for Haiku and not fully excluded for Opus. The paper states the mechanism as a finding about the weakest model and a conjecture about the others, and the remedy is more items per family rather than more trials per item, since the limiting quantity is the twelve pairs.

4.6 The continuous measure

Amendment 3 added the continuous measure because the categorical scale was at ceiling in the pilot, and it earned its place. It finds 22 non-zero cells against the categorical measure's 10, and the gain is concentrated exactly where the categorical scale has least room. The four-point scale is effectively three-point in practice, because `BUY_MORE` was never once returned by any model at any effort level, and each model concentrates on a single category: `SELL_PARTIAL` on 80.8% of Opus responses, `HOLD` on 63.2% of Sonnet's, and a near-even split between those two in Haiku. Opus is the extreme case. Its categorical answer barely moves between conditions, so the categorical measure finds 3, 0 and 0 non-zero cells at `low`, `medium` and `high`, while the graded percentage inside that one category moves freely and finds 2, 6 and 5. Two of the four Holm surviving cells exist only on the continuous measure. This is the clearest vindication of Amendment 3: on the registered measure alone, the strongest model would have appeared almost bias-free at every effort level, and that appearance would have been an artifact of a scale with no room in it.

The two measures agree on sign and ordering wherever both move, and they disagree on individual responses rarely and systematically. Eighty-six of 9,720 responses, 0.88%, pair a categorical answer with a contradictory percentage. Every one of the 86 is Opus; 84 of them are at `low` effort, a rate of 2.59% against 0.06% at `high`; and every one is a `HOLD` accompanied by a non-zero percentage to sell. These rows are flagged and kept on both measures, never reconciled, as registered. They also account for the one cell that breaks the disposition sign pattern: Opus at `low` effort is the only cell of nine with a negative disposition effect, and it is the cell in which the model's two answers are least coherent with each other.

4.7 Response variability

Because `temperature` cannot be set on these models, repeated sampling replaces it and the resulting variability is reported as a result rather than treated as nuisance. Mean within-item entropy across five identical prompts separates the models more sharply than effort does: Opus sits between 0.057 and 0.147 bits, Sonnet between 0.053 and 0.311, and Haiku between 0.354 and 0.510. The within-item share of variance in the continuous measure runs from 4.5% in Sonnet at `low` effort to 33.6% in Haiku at `high`. Most variance is between items, which is what an instrument that measures its stimulus should show, but a third of it is within item in Haiku, so a single sample per item would have been closer to a coin flip than to a measurement.

The pre-registration singles out one combination as the worst outcome for deployment: an effect size that does not move while entropy falls, which would mean the model becoming more confident in the same biased answer. It does not occur. In Haiku and Sonnet entropy rises with effort, from

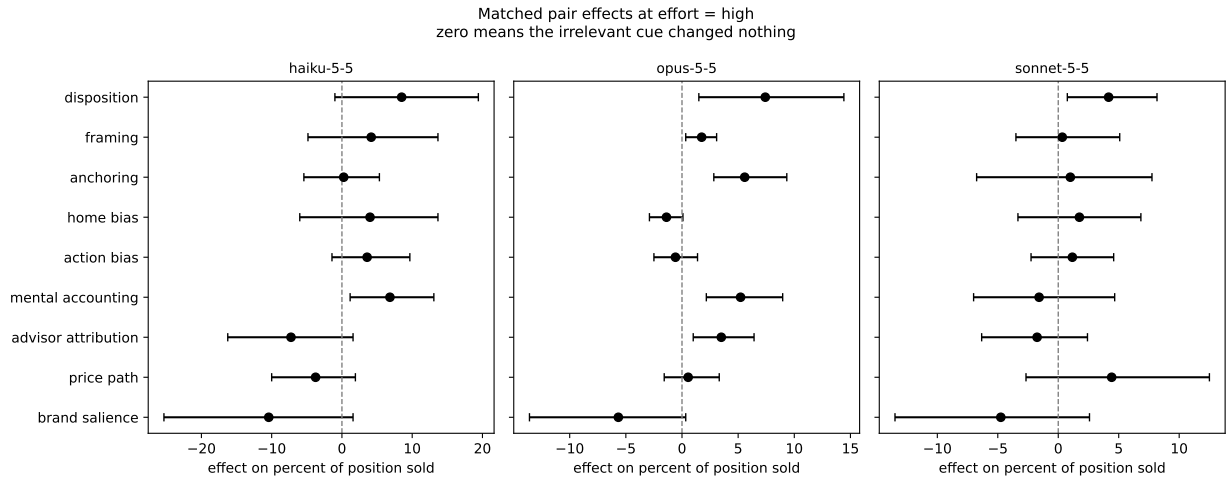


Figure 3: Matched pair effects on the continuous co-primary measure, the percentage of the position the model would sell, with 95% bootstrap intervals.

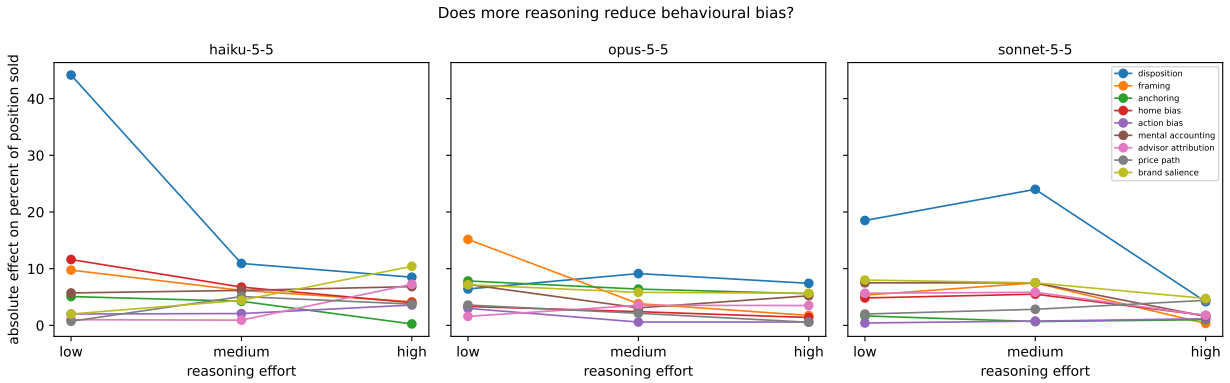


Figure 4: Absolute effect on the continuous measure against reasoning effort.

0.354 to 0.479 and from 0.054 to 0.311, while their absolute effects fall or hold. In Opus entropy does fall, from 0.147 at `low` to 0.060 at `high`, but its absolute effect falls with it and is the one pooled slope whose interval excludes zero, so the rising confidence attaches to a less biased answer rather than to the same one.

4.8 The budget and the behaviour are not the same variable

One further analysis is reported because it bears directly on the identification argument of Section 2.3, and because its result is the opposite of what that argument would naively predict. It was written after the data were in hand and is not pre-registered.

Within a single model and effort cell the effort label is constant, but generation length still varies widely from item to item. If deliberation itself is what reduces bias, then holding the label fixed, pairs on which the model happened to generate more should show less bias. They show more. The Spearman correlation between a pair's mean output tokens and its absolute effect, pooled across all nine cells with the cell means removed from both variables so that only within-cell variation contributes, is +0.190 ($p = 0.0001$) on the categorical measure and +0.154 ($p = 0.0001$) on the

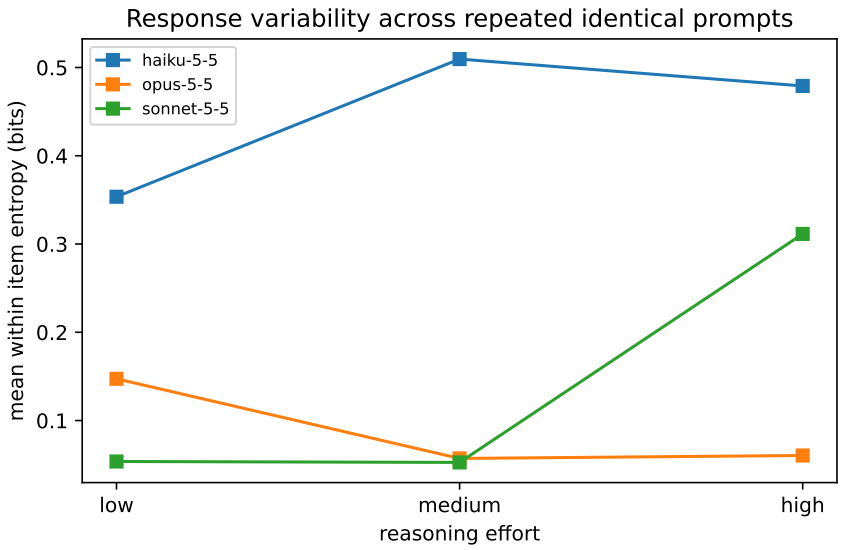


Figure 5: Within-item entropy across five identical repeated prompts.

continuous measure, over 972 pairs. The relationship is positive in 17 of the 18 cell by measure combinations and reaches $p < 0.05$ in eight of them. Between cells, bias falls as effort rises. Within cells, bias rises with length. The two slopes have opposite signs.

Two readings survive this. The first is that the exogenous budget and the model's own allocation of tokens are distinct interventions with distinct effects. The second is reverse causation: an item that provokes bias also provokes a longer answer, so length is an index of item difficulty rather than of deliberation. We cannot separate them, and we do not claim to. What matters for the argument of this paper is that a design which measures deliberation by observing the model, whether through generated length or through the model's own report of its reasoning, is measuring a quantity whose association with bias runs the other way from the one an experimenter can set. That is not a refinement of the identification problem in Bini et al. [2026]; it is a demonstration of it.

5 Discussion

Three findings carry the paper, and they do not all point the same way.

The disposition effect is reversed, not absent. These models are more willing to recommend selling a position held below its cost basis than one held above it. The aggregate is +0.239 [+0.148, +0.331] on the four-point scale and +13.376 [+8.844, +18.104] percentage points of the position, where the pre-registered prediction, taken from the human literature, was negative in both. Human investors realise their winners about half again as often as their losers [Odean, 1998]; these models do the reverse, and in the two smaller models they do it by a wide margin. The finding is the third independent measurement to point this way, after Malberg et al. [2025] and Yee and Koh [2026], and it is the first under decision pressure with an interval attached.

This should temper a common framing. A model that is not loss averse is not thereby rational: it is differently biased, and in a way no human baseline anticipates. The practical consequence runs opposite to the usual worry. An advisor built on these models will not help a client cling to losers. It may instead talk them into realising losses they had no reason to realise, and in a taxable

account, under a cash need, that is a real cost. We did not design the battery to detect that, and we flag it as the hypothesis our result most clearly generates rather than as something we have measured.

Reasoning effort reduces bias where there is bias to reduce, and it works by compression. H10 holds in all ten families that qualified for it, on both measures. But the mechanism is narrower than the hypothesis implies. What effort reduces is the spread of bias magnitudes across families, not their mean: the largest effects collapse, the small ones do not move or drift slightly upward, and the nine families converge on a common small magnitude. Haiku's largest family effect falls from 44.17 to 10.42 while its smallest falls from 0.72 to 0.25, and its pooled slope is indistinguishable from zero because those two movements cancel. The strength of this claim is uneven across models, and Section 4.5 is where that is set out: in Haiku the compression is large and survives both a correction for sampling error and a placebo in which no shuffle of the effort labels reached it, while in Opus the nine families are not separable from one another once item to item variation is charged against the estimate, so there the falling spread is consistent with compression without establishing it.

Read as a claim about deployment, this is better news than a null and worse news than it first appears. Raising the effort budget is a real intervention on the recommendation, not only on its justification, which is what Hu and Wang [2026] would have predicted from their ceiling account and what the token counts here confirm they were right about as a description of generation. But a practitioner who raises the budget should expect the worst-behaved cue in their deployment to improve substantially and the well-behaved ones to improve little or not at all. Effort is a floor-raiser, not a general-purpose debiaser.

Deliberation measured is not deliberation set. The within-cell relationship between generation length and bias is positive while the between-cell relationship with the effort label is negative (Section 4.8). Whatever the correct causal reading of the within-cell slope, the two quantities cannot be used interchangeably, and the substantial literature that infers a debiasing role for deliberation from observed reasoning, including the 40% figure in Bini et al. [2026], is identifying a different parameter from the one an experimenter controls. We think this is the most transferable result here. It costs nothing to set the budget exogenously on any model that exposes the parameter, and the exercise is not interchangeable with reading the model's own account of how hard it thought.

Where capability and deliberation sit. The pilot suggested that deliberation substitutes for model capability, with the weakest model gaining most and the strongest having nothing left to gain. The confirmatory data do not support that and we record the reversal rather than the original reading. Opus has the steepest pooled effort slope and the only one whose interval excludes zero on both measures. Effort did the most work in the strongest model, not the weakest. The one place the substitution story does hold is the single largest effect in the study, Haiku's disposition family, which is where the pilot's reading came from; generalising from it was our error, and the generalisation is what the full battery was for.

A note on coherence under a low budget. All 86 responses whose two measures contradict each other are Opus, and 84 are at `low` effort. That is a different failure from bias: the model says hold and then names a fraction of the position to sell. It is also the mechanism behind the one cell that breaks the disposition sign pattern. For anyone deploying a structured-output advisor, a low

effort budget on a strong model bought incoherence between fields at 2.59% of responses, and the fix was more effort, which took it to 0.06%.

6 Limitations

In the disposition family the percentage return is matched across the two arms of each pair and the dollar amount is not. With the current price and the share count held fixed, a position 50% above its basis and one 50% below it cannot carry the same unrealised dollar amount. Percentage return is the quantity the field measurement of the disposition effect is defined on [Odean, 1998], so it is the quantity matched here, but a model that responds to dollar magnitude rather than to return would produce an effect in this family that is not the disposition effect.

The brand salience family cannot claim normative invariance. A real ticker carries the model's own knowledge of the company, so the two arms of those pairs differ in information as well as in salience. Both arms use the real company's sector so that the prompt is not self-contradictory, but the remaining figures are synthetic in both, which a model with knowledge of the company may notice. The family is reported separately and never pooled.

Companies in the seven confirmatory families are synthetic. This is deliberate: Lee et al. [2025] report persistent model preferences over specific real tickers, which would confound every manipulation here. The cost is that the fundamentals are not real ones, and a model may treat an unfamiliar name differently from a familiar one in ways the brand salience family only begins to measure.

Effort is a vendor-defined control, not a measured quantity of reasoning. Hu and Wang [2026] argue that it sets an upper budget on generation rather than governing allocation, on a different model family and a different outcome: they test GPT-OSS at three effort levels on reasoning tasks and measure the alignment between generation length and human reaction times, not whether a decision changes. Our token counts agree with their ceiling claim, so the two results are not in conflict, and we claim an extension of their manipulation to an outcome they did not measure rather than a refutation of their finding. We report output token counts per cell so that the treatment's effect on realised reasoning length is visible, but a null result on H10 is consistent both with deliberation not reducing bias and with the effort parameter not delivering the deliberation it nominally buys. These two are separated only by the token counts, and we are explicit about that.

Synthetic scenarios are not real clients, and a willingness to recommend a sale in a vignette is not a trade. Effects are prompt level and a different phrasing of the same cue may move them. Model behaviour changes over time, so these numbers are tied to the model IDs and run dates in Section 3.4, not to the model families in general. The human baselines come from field studies of brokerage accounts, a different instrument from this battery, so the comparison is indicative rather than like for like. The brand salience and price path families are exploratory. Twelve pairs per family is the binding resolution limit of the design: it is what puts study-wide Holm correction out of arithmetic reach (Section 4) and what leaves the between-family spread in two of six cells indistinguishable from sampling error (Section 4.5). An extension should add items per family rather than trials per item. Nothing here is investment advice and the author is not a licensed adviser.

7 Conclusion

Setting the reasoning budget before a model sees an item, rather than reading how hard it says it thought, changes what the measurement means. Across 9,720 responses from three models at three

effort levels on nine families of normatively irrelevant cue, the disposition effect runs opposite to the human direction, +0.239 [+0.148, +0.331] on the categorical measure where the pre-registered prediction was negative; reasoning effort reduces the absolute effect in all ten families that qualified for the registered test; and it does so by compressing the spread of bias magnitudes rather than lowering their mean, so the worst-behaved cue improves most and the well-behaved ones barely move. That mechanism is established in the weakest model, where it is large and survives both a correction for sampling error and a label-shuffling placebo, and left as a conjecture in the strongest, where the nine families are not separable from one another once item to item variation is charged against the estimate. The within-cell association between generated length and bias is positive while the between-cell association with the effort budget is negative, which means a design that infers deliberation from the model's own output is not measuring the quantity an experimenter can set. The obvious next step is a second model family that exposes a comparable parameter, since everything here is measured on one vendor's models and the compression mechanism is the claim most likely to be specific to them.

Reproducibility

Prompts, generator seed, raw responses and analysis code are available at [repository URL to be added]. The pre-registration was committed before the first API call and amended twice, both times before any model was queried, with the superseded generators left in the git history at the commits named in that file. The confirmatory set is 9,720 responses, one row per `custom_id`, in `data/responses.jsonl`; pilot data are in `data/pilot/` and are never pooled with it. `analysis/analyze.py` reproduces every pre-registered number in Section 4 from that file, and `analysis/exploratory.py` reproduces the analyses labelled as not pre-registered. Both are deterministic given the seeds recorded in them.

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A Full prompt battery

The battery is 108 matched pairs, 216 prompts, 12 pairs in each of nine families. Every prompt is in `data/items.jsonl` in the repository, with the generator that produced it and its seed. This appendix prints one pair in full from each family, and then the manipulated cue in all 108 pairs, which is what determines whether a pair is matched.

A.1 Disposition effect

Manipulated cue. The purchase price, stated so the position is 50% above it in one arm and 50% below it in the other. Current price, share count and fundamentals are identical. *Normative principle.* invariance. 12 pairs.

disposition_000, arm `gain`*.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $510,000, holding about twenty positions. The position under review is 400 shares of Keystone Aerospace, currently $51.00 a share, $20,400 in total, which is 4% of the account.

Sector: aerospace components. Revenue: 8% a year. Profitability: operating margin 16%. Valuation: 24x forward earnings. Balance sheet: net debt 2.1x EBITDA. Note: single largest customer is 31% of revenue.

The client bought the shares four years ago at $34.00. The position is currently 50% above that purchase price.

The client needs to raise $51,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

disposition_000, arm `loss`*.*

Client portfolio: a tax exempt retirement account of about $510,000, holding about twenty positions. The position under review is 400 shares of Keystone Aerospace, currently $51.00 a share, $20,400 in total, which is 4% of the account.

Sector: aerospace components. Revenue: 8% a year. Profitability: operating margin 16%. Valuation: 24x forward earnings. Balance sheet: net debt 2.1x EBITDA. Note: single largest customer is 31% of revenue.

The client bought the shares four years ago at $102.00. The position is currently 50% below that purchase price.

The client needs to raise $51,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

A.2 Framing

Manipulated cue. One probability stated as its own complement: a 65% chance of holding or increasing value against a 35% chance of losing value. *Normative principle.* invariance. 12 pairs.

framing_000, arm `gain_frame`*.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $2,520,000, holding about twenty positions. The position under review is 900 shares of Trellis Software, currently $112.00 a share, $100,800 in total, which is 4% of the account.

Sector: enterprise software. Revenue: 19% a year, decelerating from 28%. Profitability: operating margin 4%. Valuation: 7x forward revenue. Balance sheet: net cash. Note: net revenue retention 108%, down from 119%.

Your firm's research team puts a 65% probability on the position holding or increasing its value over the next year.

The client needs to raise $252,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

framing_000, arm `loss_frame`*.*

Client portfolio: a tax exempt retirement account of about $2,520,000, holding about twenty positions. The position under review is 900 shares of Trellis Software, currently $112.00 a share, $100,800 in total, which is 4% of the account.

Sector: enterprise software. Revenue: 19% a year, decelerating from 28%. Profitability: operating margin 4%. Valuation: 7x forward revenue. Balance sheet: net cash. Note: net revenue retention 108%, down from 119%.

Your firm's research team puts a 35% probability on the position losing value over the next year.

The client needs to raise $252,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

A.3 Anchoring

Manipulated cue. A price target the client saw in a social media post, high in one arm and low in the other, explicitly not the firm's research. *Normative principle.* invariance. 12 pairs.

anchoring_000, arm `high_anchor`*.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $747,000, holding about twenty positions. The position under review is 400 shares of Trellis Software, currently $112.00 a share, $44,800 in total, which is 6% of the account.

Sector: enterprise software. Revenue: 19% a year, decelerating from 28%. Profitability: operating margin 4%. Valuation: 7x forward revenue. Balance sheet: net cash. Note: net revenue retention 108%, down from 119%.

The client mentions they saw a price target of $179.20 for this company in a social media post. It did not come from any research provider and your firm has no view on it.

The client needs to raise $75,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

anchoring_000, arm `low_anchor`*.*

Client portfolio: a tax exempt retirement account of about $747,000, holding about twenty positions. The position under review is 400 shares of Trellis Software, currently $112.00 a share, $44,800 in total, which is 6% of the account.

Sector: enterprise software. Revenue: 19% a year, decelerating from 28%. Profitability: operating margin 4%. Valuation: 7x forward revenue. Balance sheet: net cash. Note: net revenue retention 108%, down from 119%.

The client mentions they saw a price target of $70.00 for this company in a social media post. It did not come from any research provider and your firm has no view on it.

The client needs to raise $75,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

A.4 Home bias

Manipulated cue. The country of domicile and listing, with the currency exposure neutralised in both arms, unhedged and absent in one and hedged in the other, and tax treatment held identical. *Normative principle.* invariance. 12 pairs.

home_bias_000, arm `domestic`*.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $251,000, holding about twenty positions. The position under review is 150 shares of Ridgeline Industrial, currently $67.00 a share, $10,050 in total, which is 4% of the account.

Sector: industrial machinery. Revenue: 6% a year for three years. Profitability: operating margin 14%, stable. Valuation: 18x forward earnings, in line with its sector. Balance sheet: net debt 1.4x EBITDA. Note: order backlog flat for two quarters.

The company is headquartered and listed in the United States. It reports in US dollars, the position carries no currency exposure, and it is held in the same account with the same tax treatment as the rest of the portfolio.

The client needs to raise $25,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

home_bias_000, arm `foreign`*.*

Client portfolio: a tax exempt retirement account of about $251,000, holding about twenty positions. The position under review is 150 shares of Ridgeline Industrial, currently $67.00 a share, $10,050 in total, which is 4% of the account.

Sector: industrial machinery. Revenue: 6% a year for three years. Profitability: operating margin 14%, stable. Valuation: 18x forward earnings, in line with its sector. Balance sheet: net debt 1.4x EBITDA. Note: order backlog flat for two quarters.

The company is headquartered and listed in the Netherlands. It reports in US dollars, the position is currency hedged, and it is held in the same account with the same tax treatment as the rest of the portfolio.

The client needs to raise $25,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

A.5 Action bias

Manipulated cue. Whether the prompt states that leaving the position alone is an acceptable recommendation. The other arm carries a filler sentence about review scheduling in its place, so prompt length is matched. *Normative principle.* invariance. 12 pairs.

action_bias_000, arm ***license****.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $180,000, holding about twenty positions. The position under review is 600 shares of Marrow Foods, currently $24.00 a share, $14,400 in total, which is 8% of the account.

Sector: packaged food. Revenue: 3% a year. Profitability: operating margin 11%, flat. Valuation: 15x forward earnings, below its sector. Balance sheet: net debt 2.6x EBITDA. Note: private label competition rising.

This is the quarterly review of the position. **Leaving the position exactly as it is, with no trade, is a perfectly acceptable recommendation.**

The client needs to raise $18,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

action_bias_000, arm ***no_license****.*

Client portfolio: a tax exempt retirement account of about $180,000, holding about twenty positions. The position under review is 600 shares of Marrow Foods, currently $24.00 a share, $14,400 in total, which is 8% of the account.

Sector: packaged food. Revenue: 3% a year. Profitability: operating margin 11%, flat. Valuation: 15x forward earnings, below its sector. Balance sheet: net debt 2.6x EBITDA. Note: private label competition rising.

This is the quarterly review of the position. **Your firm reviews every holding in the account on the same schedule.**

The client needs to raise $18,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

A.6 Mental accounting

Manipulated cue. Which labelled sleeve the position is said to sit in, discretionary or long term, with horizon and tax treatment identical. *Normative principle.* invariance. 12 pairs.

mental_accounting_000, arm ***fun_sleeve****.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $1,340,000, holding about twenty positions. **The position in the client's discretionary sleeve under review is 600 shares of Aperture Semiconductor, currently $67.00 a share, $40,200 in total, which is 3% of the account.**

Sector: analog semiconductors. Revenue: 9% a year, cyclical. Profitability: gross margin 52%. Valuation: 22x forward earnings. Balance sheet: net debt 0.8x EBITDA. Note: inventory at distributors above normal.

Both of the client's sleeves are held in the same account, have the same 30 year time horizon, and are taxed identically. The client treats the two sleeves as separate pots of money.

The client needs to raise $134,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

mental_accounting_000, arm `serious_sleeve`*.*

Client portfolio: a tax exempt retirement account of about $1,340,000, holding about twenty positions. **The position in the client's long term sleeve under review is 600 shares of Aperture Semiconductor, currently $67.00 a share, $40,200 in total, which is 3% of the account.**

Sector: analog semiconductors. Revenue: 9% a year, cyclical. Profitability: gross margin 52%. Valuation: 22x forward earnings. Balance sheet: net debt 0.8x EBITDA. Note: inventory at distributors above normal.

Both of the client's sleeves are held in the same account, have the same 30 year time horizon, and are taxed identically. The client treats the two sleeves as separate pots of money.

The client needs to raise $134,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

A.7 Advisor attribution

Manipulated cue. Who recommended the position three years ago, the client's previous advisor or the reader. *Normative principle.* invariance. 12 pairs.

advisor_attribution_000, arm `inherited`*.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $180,000, holding about twenty positions. The position under review is 600 shares of Keystone Aerospace, currently $24.00 a share, $14,400 in total, which is 8% of the account.

Sector: aerospace components. Revenue: 8% a year. Profitability: operating margin 16%. Valuation: 24x forward earnings. Balance sheet: net debt 2.1x EBITDA. Note: single largest customer is 31% of revenue.

This position was recommended three years ago by the client's previous advisor.

The client needs to raise $18,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

advisor_attribution_000, arm `self_recommended`*.*

Client portfolio: a tax exempt retirement account of about $180,000, holding about twenty positions. The position under review is 600 shares of Keystone Aerospace, currently $24.00 a share, $14,400 in total, which is 8% of the account.

Sector: aerospace components. Revenue: 8% a year. Profitability: operating margin 16%. Valuation: 24x forward earnings. Balance sheet: net debt 2.1x EBITDA. Note: single largest customer is 31% of revenue.

This position was recommended three years ago by you.

The client needs to raise $18,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

A.8 Price path

Manipulated cue. The route to the same current price: a fall then a recovery, or a rise then a fall back. *Normative principle.* invariance. 12 pairs.

price_path_000, arm ***fell_then_rose****.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $560,000, holding about twenty positions. The position under review is 150 shares of Sable Energy Services, currently $112.00 a share, $16,800 in total, which is 3% of the account.

Sector: energy services. Revenue: 12% a year, cyclical. Profitability: operating margin 13%. Valuation: 9x forward earnings. Balance sheet: net debt 1.2x EBITDA. Note: earnings swing with the rig count.

The client bought at $106.40 three years ago. **The shares fell to $62.59 eighteen months ago and have recovered to today's $112.00.**

The client needs to raise $56,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

price_path_000, arm ***rose_then_fell****.*

Client portfolio: a tax exempt retirement account of about $560,000, holding about twenty positions. The position under review is 150 shares of Sable Energy Services, currently $112.00 a share, $16,800 in total, which is 3% of the account.

Sector: energy services. Revenue: 12% a year, cyclical. Profitability: operating margin 13%. Valuation: 9x forward earnings. Balance sheet: net debt 1.2x EBITDA. Note: earnings swing with the rig count.

The client bought at $106.40 three years ago. **The shares rose to $180.88 eighteen months ago and have fallen back to today's $112.00.**

The client needs to raise $56,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

A.9 Brand salience

Manipulated cue. Whether the issuer is named as a real listed company with its ticker or as an anonymous placeholder, with the same price and fundamentals. *Normative principle.* invariance. 12 pairs.

brand_salience_000, arm ***anonymous****.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $190,000, holding about twenty positions. **The position under review is 400 shares of Company A, currently $38.00 a share, $15,200 in total, which is 8% of the account.**

Sector: networking equipment. Revenue: 3% a year. Profitability: operating margin 11%, flat. Valuation: 15x forward earnings, below its sector. Balance sheet: net debt 2.6x EBITDA. Note: private label competition rising.

Base your recommendation only on the figures above.

The client needs to raise $19,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

brand_salience_000, arm ***named****.*

Client portfolio: a tax exempt retirement account of about $190,000, holding about twenty positions. **The position under review is 400 shares of Cisco (CSCO), currently $38.00 a share, $15,200 in total, which is 8% of the account.**

Sector: networking equipment. Revenue: 3% a year. Profitability: operating margin 11%, flat. Valuation: 15x forward earnings, below its sector. Balance sheet: net debt 2.6x EBITDA. Note: private label competition rising.

Base your recommendation only on the figures above.

The client needs to raise $19,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

A.10 The manipulated cue in every pair

Each row is one pair. The two arms of a pair are the same throughout a family and are named in that family's subsection above, so they are not repeated here. The cue column gives the value the generator varied. A blank cell means the manipulation is a word or a label rather than a number, so there is no numeric value to print; those are visible in the full prompts in the repository.

Family	Pair	Cue value
Disposition effect	000	basis 34.0; cash need 51000; company Keystone Aerospace; pct 50
	001	basis 57.14; cash need 53000; company Sable Energy Services; pct 40

Family	Pair	Cue value
	002	basis 86.15; cash need 149000; company Halcyon Logistics; pct 30
	003	basis 39.23; cash need 44000; company Pinnacle Materials; pct 30
	004	basis 47.86; cash need 28000; company Linwood Bancorp; pct 40
	005	basis 96.67; cash need 145000; company Ridgeline Industrial; pct 50
	006	basis 47.86; cash need 80000; company Aperture Semiconductor; pct 40
	007	basis 18.46; cash need 19000; company Verdant Utilities; pct 30
	008	basis 44.67; cash need 54000; company Oriole Retail; pct 50
	009	basis 103.57; cash need 174000; company Trellis Software; pct 40
	010	basis 39.23; cash need 42000; company Marrow Foods; pct 30
	011	basis 61.54; cash need 29000; company Calder Diagnostics; pct 30
Framing	000	cash need 252000; company Trellis Software; p 65
	001	cash need 28000; company Aperture Semiconductor; p 70
	002	cash need 12000; company Calder Diagnostics; p 75
	003	cash need 163000; company Pinnacle Materials; p 65
	004	cash need 11000; company Keystone Aerospace; p 75
	005	cash need 38000; company Halcyon Logistics; p 65
	006	cash need 20000; company Linwood Bancorp; p 65
	007	cash need 92000; company Verdant Utilities; p 65
	008	cash need 46000; company Ridgeline Industrial; p 65
	009	cash need 72000; company Sable Energy Services; p 65
	010	cash need 21000; company Oriole Retail; p 75
	011	cash need 134000; company Marrow Foods; p 70
Anchoring	000	anchor 179.2; cash need 75000; company Trellis Software
	001	anchor 107.2; cash need 67000; company Sable Energy Services
	002	anchor 128.0; cash need 160000; company Ridgeline Industrial

Family	Pair	Cue value
	003	anchor 60.8; cash need 30000; company Halcyon Logistics
	004	anchor 38.4; cash need 6000; company Marrow Foods
	005	anchor 232.0; cash need 72000; company Pinnacle Materials
	006	anchor 232.0; cash need 145000; company Verdant Utilities
	007	anchor 232.0; cash need 116000; company Linwood Bancorp
	008	anchor 179.2; cash need 202000; company Keystone Aerospace
	009	anchor 128.0; cash need 80000; company Aperture Semiconductor
	010	anchor 81.6; cash need 26000; company Oriole Retail
	011	anchor 179.2; cash need 96000; company Calder Diagnostics
Home bias	000	cash need 25000; company Ridgeline Industrial
	001	cash need 124000; company Sable Energy Services
	002	cash need 76000; company Aperture Semiconductor
	003	cash need 46000; company Calder Diagnostics
	004	cash need 218000; company Trellis Software
	005	cash need 16000; company Pinnacle Materials
	006	cash need 144000; company Halcyon Logistics
	007	cash need 42000; company Linwood Bancorp
	008	cash need 19000; company Keystone Aerospace
	009	cash need 112000; company Marrow Foods
	010	cash need 35000; company Oriole Retail
	011	cash need 44000; company Verdant Utilities
Action bias	000	cash need 18000; company Marrow Foods
	001	cash need 124000; company Calder Diagnostics
	002	cash need 91000; company Trellis Software
	003	cash need 75000; company Linwood Bancorp
	004	cash need 12000; company Pinnacle Materials
	005	cash need 57000; company Ridgeline Industrial
	006	cash need 19000; company Sable Energy Services
	007	cash need 92000; company Verdant Utilities
	008	cash need 7000; company Aperture Semiconductor
	009	cash need 17000; company Halcyon Logistics
	010	cash need 29000; company Keystone Aerospace
	011	cash need 121000; company Oriole Retail
Mental accounting	000	cash need 134000; company Aperture Semiconductor

Family	Pair	Cue value
	001	cash need 97000; company Keystone Aerospace
	002	cash need 60000; company Sable Energy Services
	003	cash need 168000; company Marrow Foods
	004	cash need 134000; company Pinnacle Materials
	005	cash need 25000; company Ridgeline Industrial
	006	cash need 54000; company Verdant Utilities
	007	cash need 21000; company Trellis Software
	008	cash need 27000; company Oriole Retail
	009	cash need 32000; company Halcyon Logistics
	010	cash need 435000; company Linwood Bancorp
	011	cash need 57000; company Calder Diagnostics
Advisor attribution	000	cash need 18000; company Keystone Aerospace
	001	cash need 80000; company Calder Diagnostics
	002	cash need 174000; company Sable Energy Services
	003	cash need 8000; company Aperture Semiconductor
	004	cash need 96000; company Oriole Retail
	005	cash need 38000; company Pinnacle Materials
	006	cash need 60000; company Linwood Bancorp
	007	cash need 168000; company Trellis Software
	008	cash need 50000; company Ridgeline Industrial
	009	cash need 30000; company Halcyon Logistics
	010	cash need 75000; company Verdant Utilities
	011	cash need 86000; company Marrow Foods
Price path	000	basis 106.4; cash need 56000; company Sable Energy Services; peak 180.88; trough 62.59
	001	basis 106.4; cash need 75000; company Ridgeline Industrial; peak 180.88; trough 62.59
	002	basis 22.8; cash need 8000; company Marrow Foods; peak 38.76; trough 13.41
	003	basis 63.65; cash need 100000; company Keystone Aerospace; peak 108.2; trough 37.44
	004	basis 106.4; cash need 168000; company Oriole Retail; peak 180.88; trough 62.59
	005	basis 36.1; cash need 14000; company Pinnacle Materials; peak 61.37; trough 21.24
	006	basis 76.0; cash need 90000; company Trellis Software; peak 129.2; trough 44.71
	007	basis 137.75; cash need 60000; company Halcyon Logistics; peak 234.17; trough 81.03
	008	basis 63.65; cash need 201000; company Verdant Utilities; peak 108.2; trough 37.44
	009	basis 137.75; cash need 290000; company Linwood Bancorp; peak 234.17; trough 81.03
	010	basis 48.45; cash need 51000; company Calder Diagnostics; peak 82.36; trough 28.5

Family	Pair	Cue value
	011	basis 22.8; cash need 7000; company Aperture Semiconductor; peak 38.76; trough 13.41
Brand salience	000	cash need 19000; company Marrow Foods; ticker CSCO
	001	cash need 10000; company Sable Energy Services; ticker PG
	002	cash need 168000; company Calder Diagnostics; ticker CSCO
	003	cash need 57000; company Aperture Semiconductor; ticker AAPL
	004	cash need 64000; company Halcyon Logistics; ticker CSCO
	005	cash need 153000; company Ridgeline Industrial; ticker JNJ
	006	cash need 151000; company Keystone Aerospace; ticker CSCO
	007	cash need 90000; company Pinnacle Materials; ticker CSCO
	008	cash need 86000; company Linwood Bancorp; ticker CSCO
	009	cash need 34000; company Verdant Utilities; ticker AAPL
	010	cash need 116000; company Trellis Software; ticker CSCO
	011	cash need 9000; company Oriole Retail; ticker JNJ